

Application programming interface

SEE ALSO

◀ 72–73 Abstraction

Python 130–131 ▶

Cloud computing 152–153 ▶

The Internet of Things 226–227 ▶

Websites often feature embedded functions such as maps or social media feeds. They do not create these themselves, but use an application programming interface (API).

What is an API?

An interface describes the way a program interacts with another. The other system can be a user, through a user interface (UI), or another program, through an API. APIs make it easier for programmers to use functions and objects from other programs in their code. When an API function is requested, the computer hosting the API executes the function's code and sends the result back to the program requesting it.

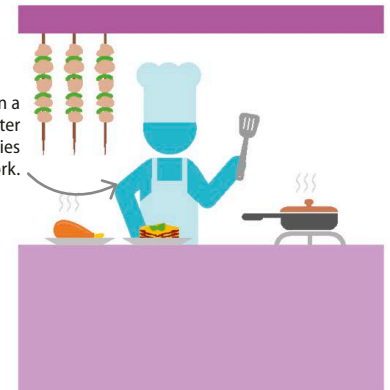
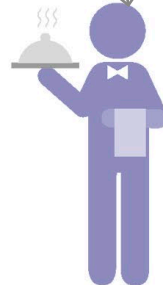
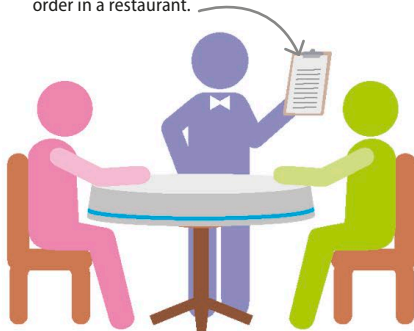
▽ Abstraction

An API is an abstraction of the program it represents. Just as a menu only lists the names of dishes and not their recipes, an API only shows the features that can be used by other programs. All details of the program's construction are hidden.

An API is like a menu of the dishes available to order in a restaurant.

The result of a request to the API is returned by the server.

Just like a chef in a restaurant, the computer hosting the API carries out the actual work.



Which languages?

APIs are written in a variety of programming languages, including PHP, Python, Ruby, and Java. Programs that are written in a different language from the one an API is written in can still use its functions. Requests to an API make use of the Hypertext Transfer Protocol (HTTP) used to transfer information across the world wide web. The API returns the result to the calling program in a standard format.

▷ Helper libraries

Many APIs provide helper libraries for different programming languages, which makes it easier to call them using another language. For example, a program written in Python can use an API's Python helper library.

